Population genomics and transcriptomics unravel the evolution of a social supergene in the ant *Myrmecina graminicola*

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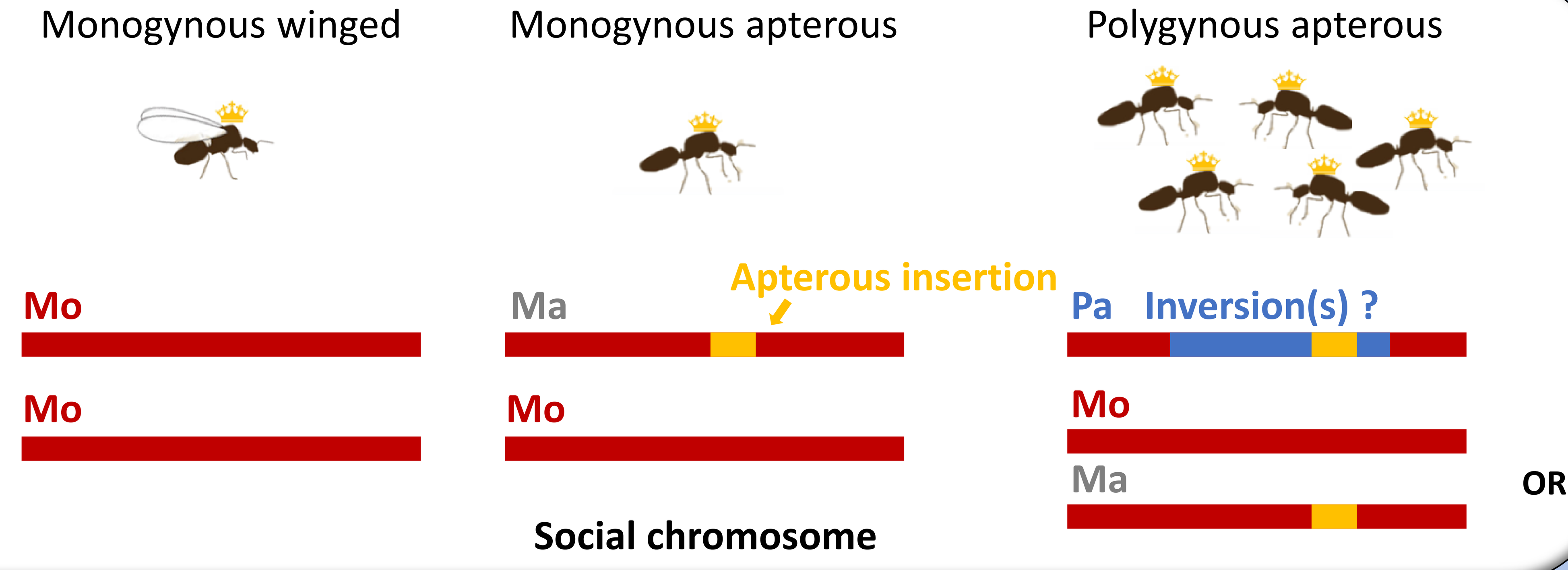
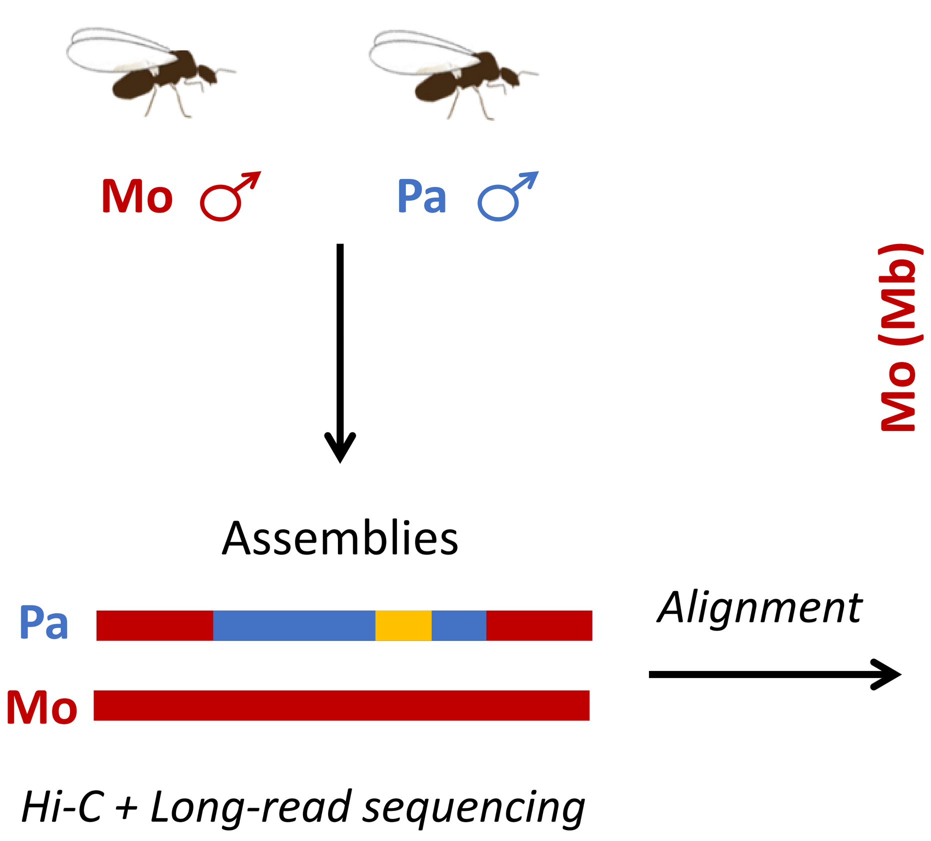
Context

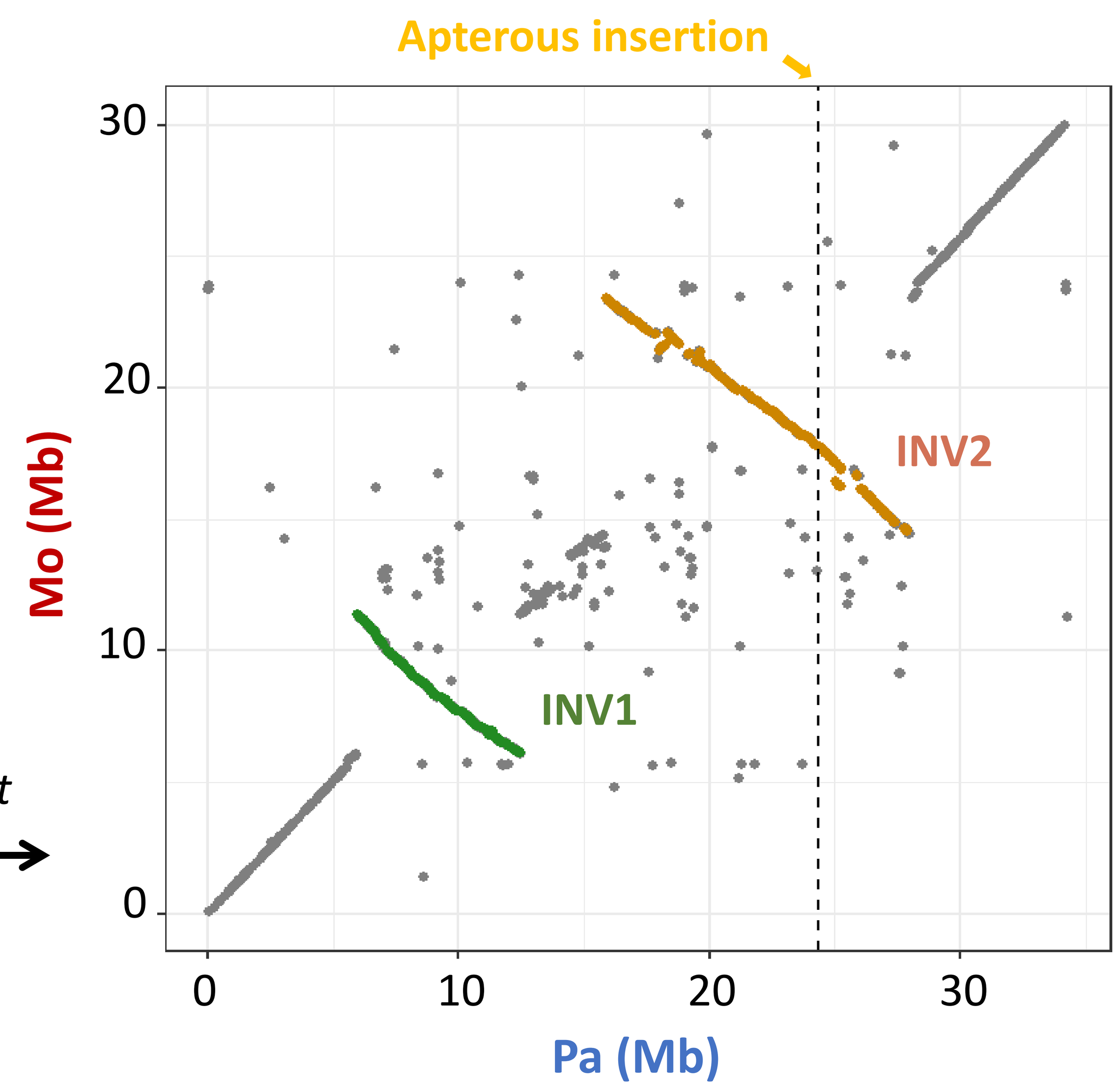
Supergenes play an important role in the evolution of complex phenotypes, locking together specific combinations of genes by suppressing recombination between haplotypes. Ultimately, this leads to the degeneration of the non-recombining haplotype. Here, we investigate the origins, the evolution and the effects of recombination suppression in the ant ***Myrmecina graminicola***, where a supergene system has been discovered [1], coding for two traits : the **number of reproductive queens** in a colony (single vs multiple queens) and the presence of **wings**.

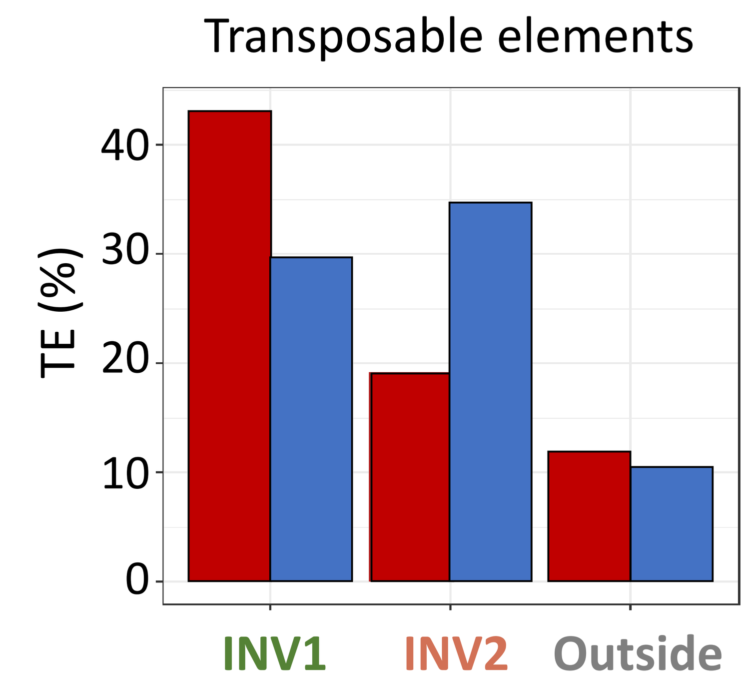


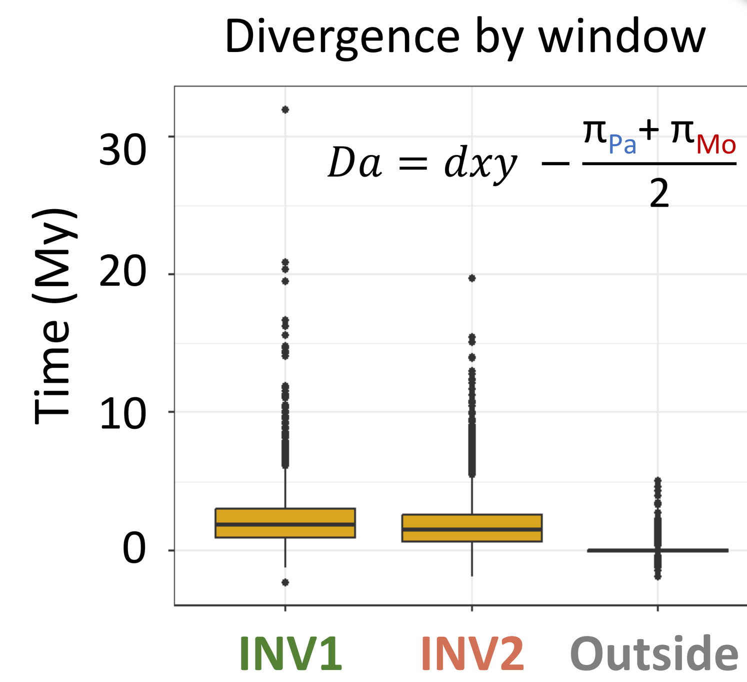
Supergene structure

How different are Mo and Pa ?





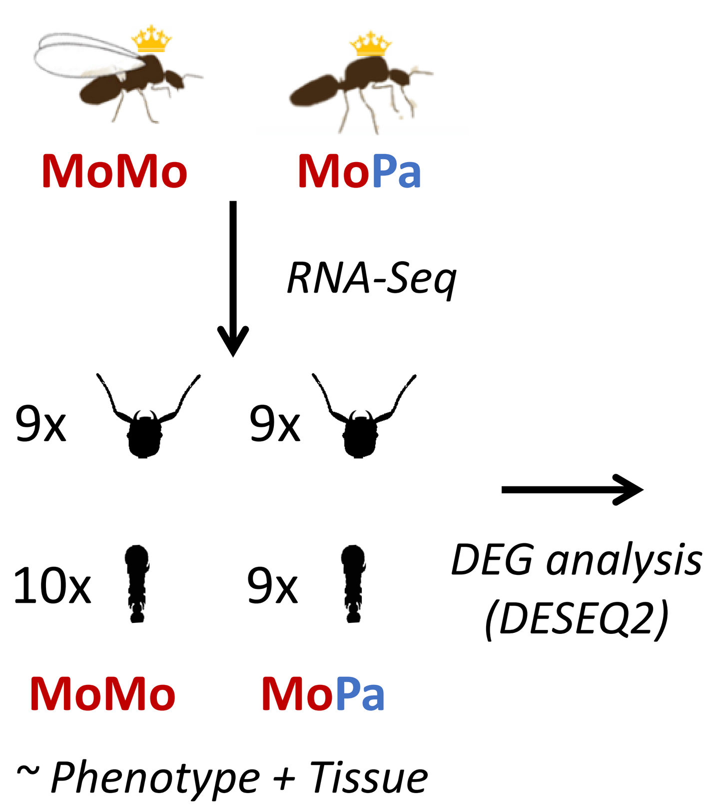


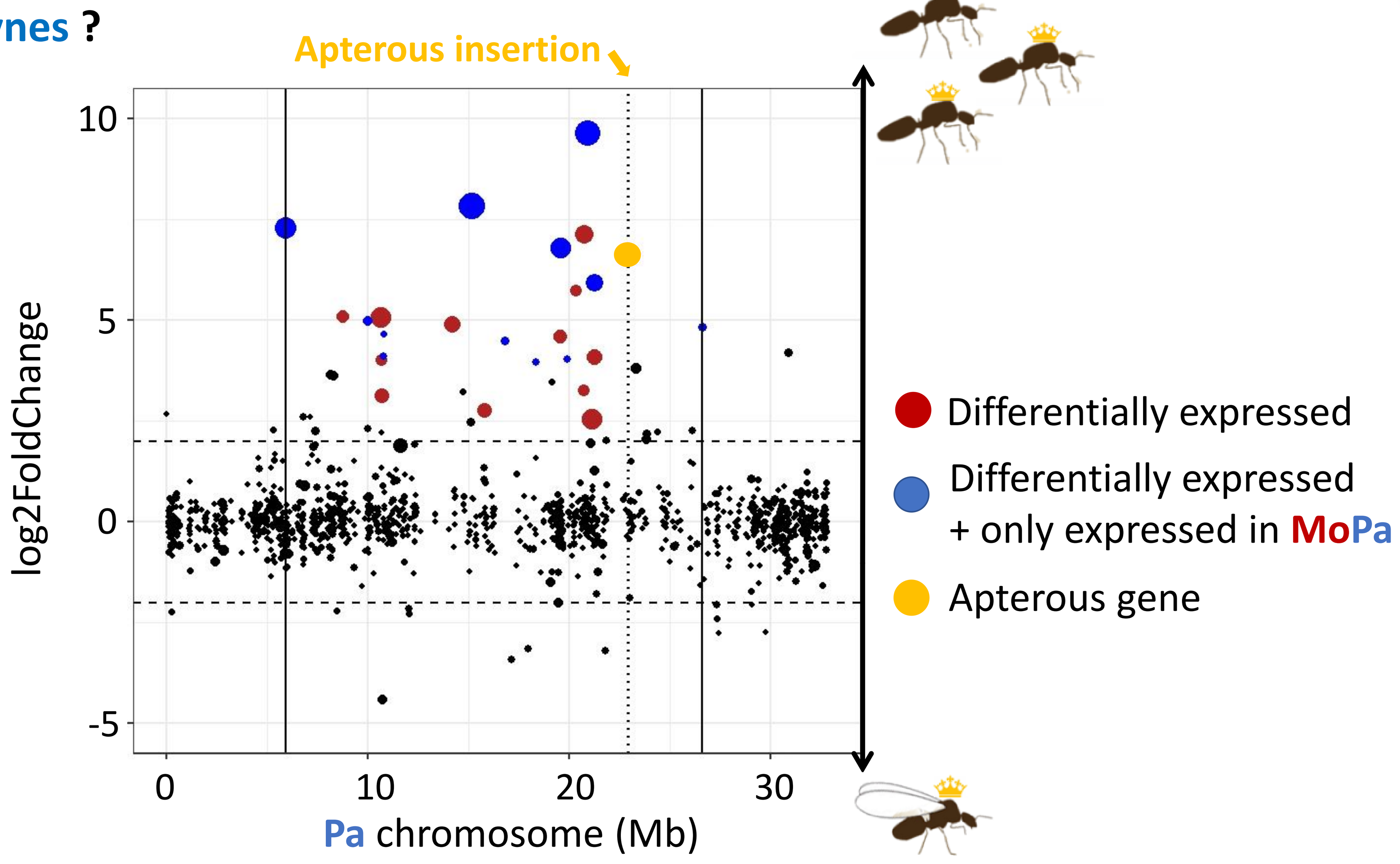


⇒ The non-recombining Pa is larger than Mo (21Mb vs 17Mb)
⇒ Two paracentric inversions INV1 and INV2
⇒ INV2 has more repeats in Pa, not INV1
⇒ Equivalent divergence time for both inversions (1.7Mya-1.8Mya)
⇒ Extremely low diversity in Pa vs Mo (π_{Pa}= 9.35x10^{-5} < π_{Mo}= 2.8x10^{-3})

Differential gene expression

How does the absence of recombination impact gene expression in polygynes ?

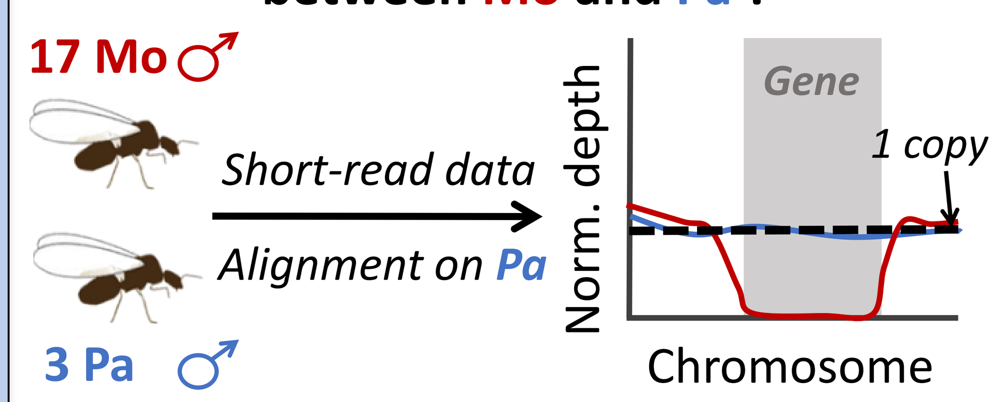




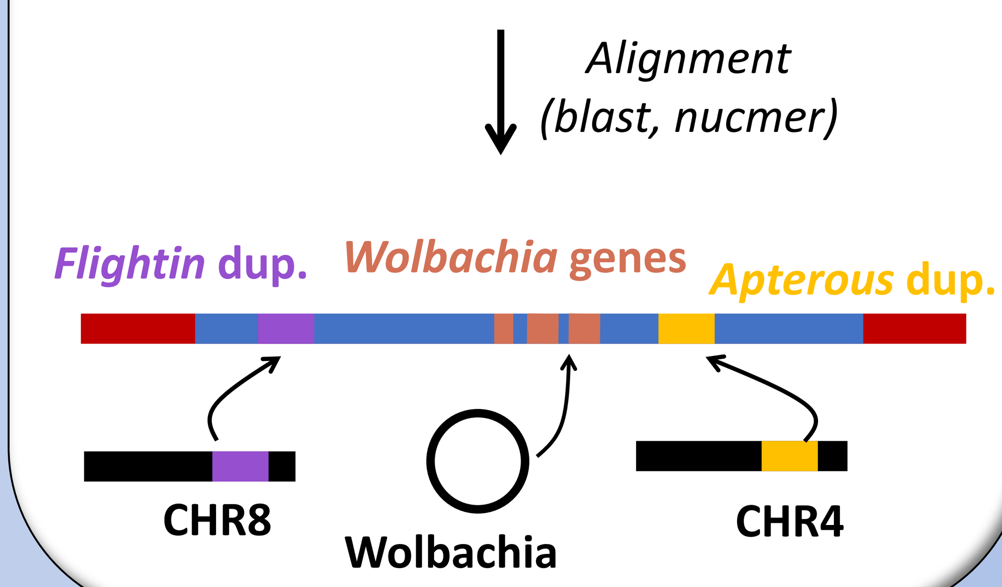
⇒ Only one gene expressed in the apterous insertion
⇒ Overexpression in polygynes
⇒ 18/24 DEG are located inside the supergene (Fisher's Exact Test p-value < 0.05)

Gene duplications

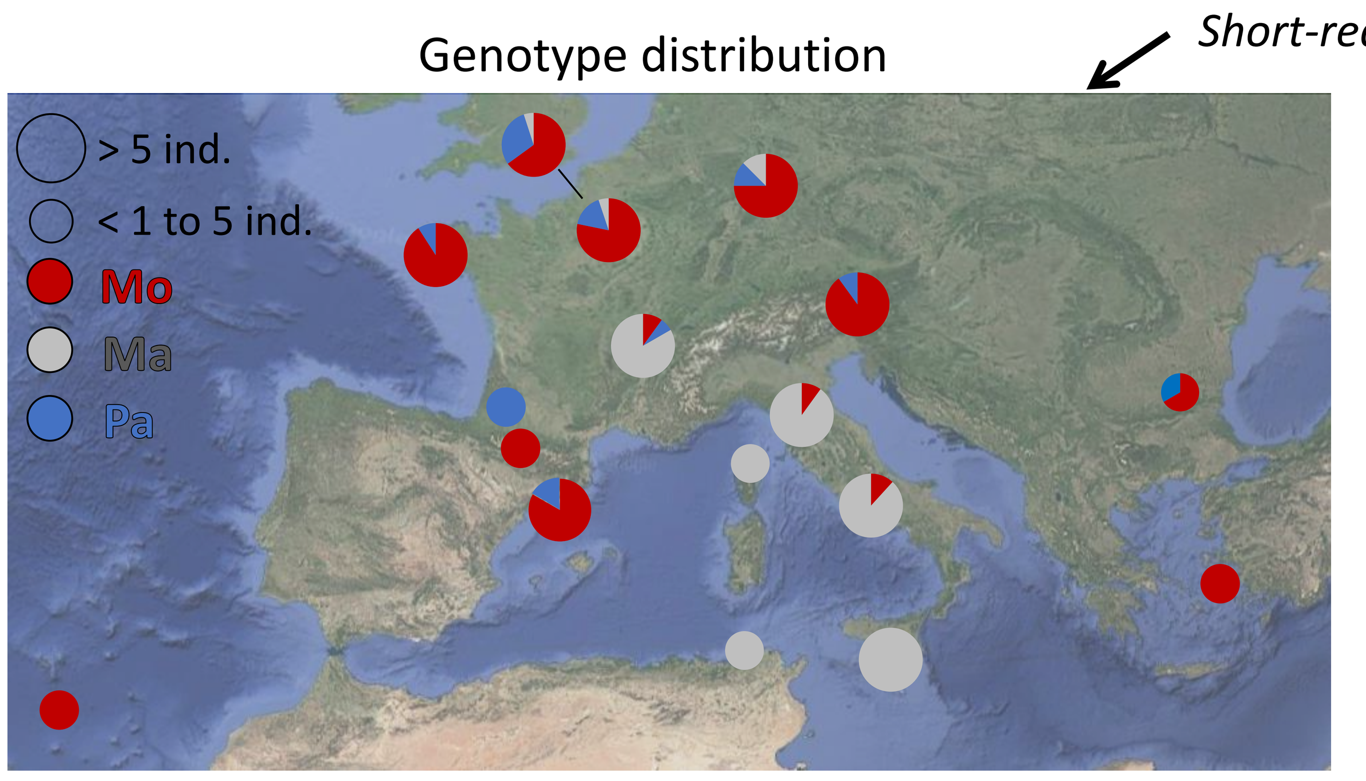
Is there gene content differences between Mo and Pa ?

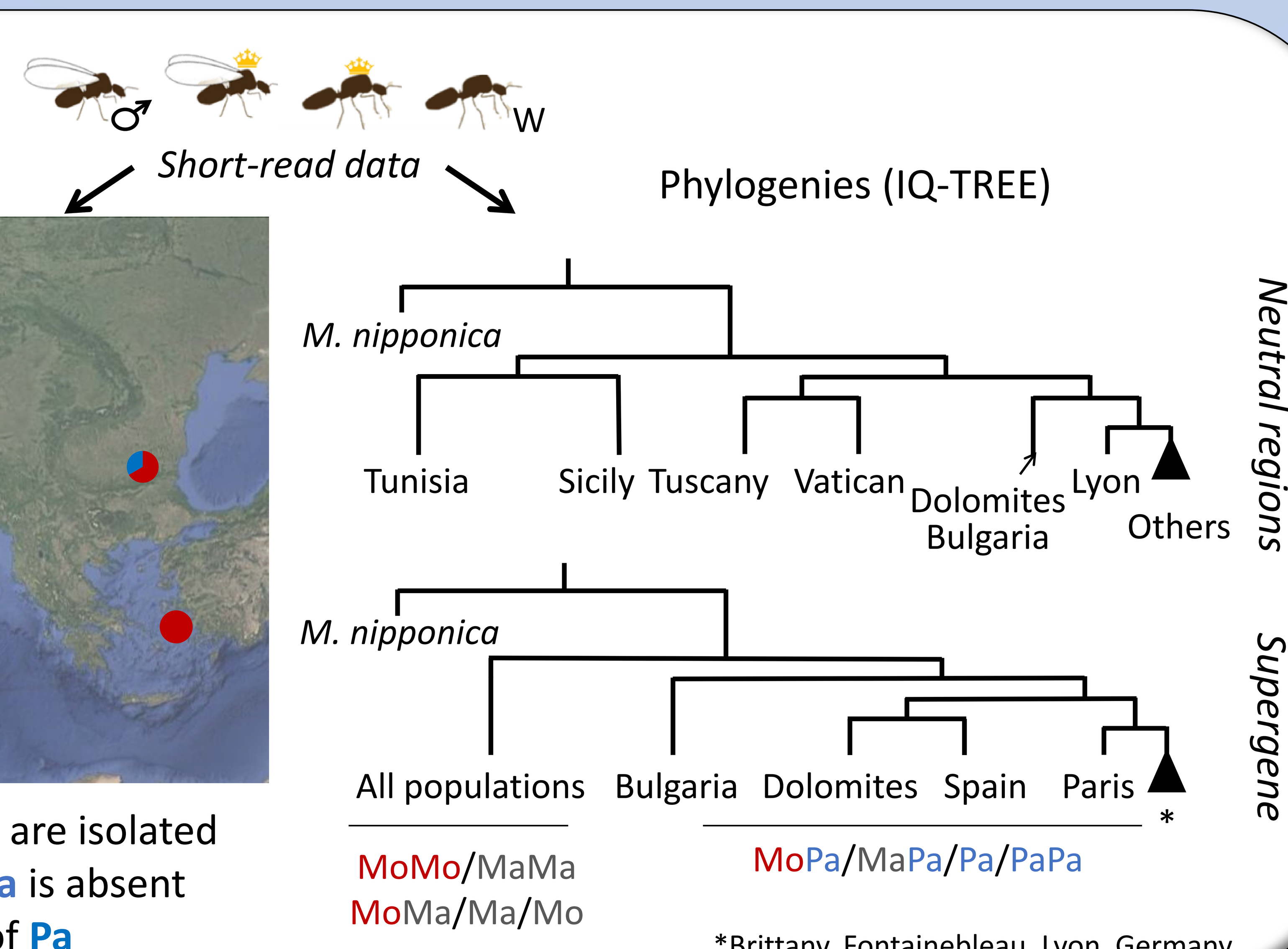


⇒ **57** genes with no depth in **Mo males**
⇒ **8** genes not expressed in **monogynes**



Phylogeography





⇒ Populations from Tunisia, Sicily and Italy are isolated
⇒ Ma predominates in the south, where Pa is absent
⇒ Five geographically structured variants of Pa

Conclusion

- Two inversions INV1 and INV2
- High divergence between Pa and Mo but no clear signs of degeneration
- Differential expression is stronger in the supergene and biased towards polygyne queens
- Pa accumulated genes from Wolbachia and autosomes
- Multiple Pa variants across Europe, absent from southern populations

References

[1] *Mona, Gay et al., Genomic evidence of a complex supergene system linking dispersal to social polymorphism, Current Biology, 2025*